

Data Science Techniques & Tools Lab Report

College Placement Analysis Dashboard
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T. Y. Artificial Intelligence and Data Science (2025-2026)



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Abstract

The College Placement Analysis Dashboard is a data analytics and visualization project developed using Microsoft Power BI. The primary objective of this project is to analyze student placement data and present meaningful insights through interactive and user-friendly visualizations.

The dashboard integrates placement-related data including student details, company information, salary packages, and placement status. It transforms raw data into structured insights using Power BI features such as Power Query for data cleaning and DAX (Data Analysis Expressions) for creating calculated measures.

Key parameters such as placement percentage, average salary, company-wise hiring, branch-wise performance, and skill-based analysis are visualized using KPI cards, bar charts, pie charts, line charts, scatter plots, and gauge charts. The dashboard is divided into multiple pages to provide detailed analysis from different perspectives.

Interactive features such as filters (slicers), drill-down, tooltips, and dynamic visuals enhance user experience and allow deeper exploration of the data. The project follows best practices in data visualization, ensuring clarity, readability, and effective communication of insights.

This dashboard serves as a practical example of how business intelligence tools can be applied to academic data for decision-making and performance analysis. It also provides a foundation for future enhancements such as predictive analytics and AI-based placement recommendations.

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1. Introduction

Overview of the Project

The College Placement Analysis Dashboard is an interactive business intelligence project developed using Microsoft Power BI. It is designed to analyze placement data and present insights related to student performance, company hiring trends, and salary distribution.

The dashboard enables users to monitor placement statistics, identify trends, and derive insights that support better decision-making for students and placement cells.

Brief Summary

The project uses placement data stored in Excel, which is transformed using Power Query and visualized using Power BI. The dashboard includes multiple pages that provide insights into:

- Student placement performance
- Company hiring patterns
- Salary distribution
- Skills impact on placement

2. Problem Statement

Placement data is often maintained in spreadsheets, making it difficult to:

- Analyze trends
- Compare performance
- Generate insights

There is a need for an interactive system that can:

- Visualize placement data
- Provide real-time insights
- Support decision-making

A **Power BI dashboard** that transforms raw data into meaningful insights.

3. Objectives

- To analyze student placement data
- To calculate placement percentage
- To identify top-performing branches
- To analyze company hiring trends
- To study salary distribution
- To visualize skill-based placement insights
- To create an interactive and user-friendly dashboard

Background

1. Tools and Technologies

The College Placement Analysis Dashboard is developed using industry-standard business intelligence and data visualization tools, selected for their ability to handle structured data, perform analytical processing, and generate interactive reports.

Power BI Desktop

The primary development environment used for designing, building, and publishing the interactive dashboard. Microsoft Power BI Desktop provides a wide range of visualization tools, data modelling features, and an integrated query editor for transforming and analyzing placement data.

Excel Dataset

Microsoft Excel is used as the primary data source for storing placement-related data, including student information, company details, and salary packages. The dataset is imported into Power BI for further processing and visualization.

Power Query Editor

Power Query Editor is used to clean, transform, and prepare the dataset before visualization. It handles tasks such as:

- Removing duplicate records
- Handling missing values
- Renaming columns
- Converting data types
- Structuring data into a usable format

DAX (Data Analysis Expressions)

DAX is used within Power BI to create calculated measures and dynamic KPIs. It enables analytical computations such as:

- Placement percentage
- Average salary
- Total students placed
- Conditional calculations for dashboard insights

Data Modeling

Data modeling is used to establish relationships between multiple tables such as:

- Students
- Companies
- Placements

This ensures accurate data analysis and efficient querying within the dashboard.

2. Software and Hardware Requirements

Software Requirements

- Microsoft Power BI Desktop (Latest Version)
- Microsoft Excel (Dataset preparation)
- Web Browser (for reference and documentation)
- Git / GitHub (for project version control – optional)

Hardware Requirements

- Computer / Laptop with minimum **4 GB RAM (8 GB recommended)**
- Minimum **Intel i3 processor or equivalent**
- Active Internet Connection (for updates and resource access)
- Display with minimum **1366 × 768 resolution** for proper dashboard viewing

4. Dashboard Overview

Architecture and Data Flow

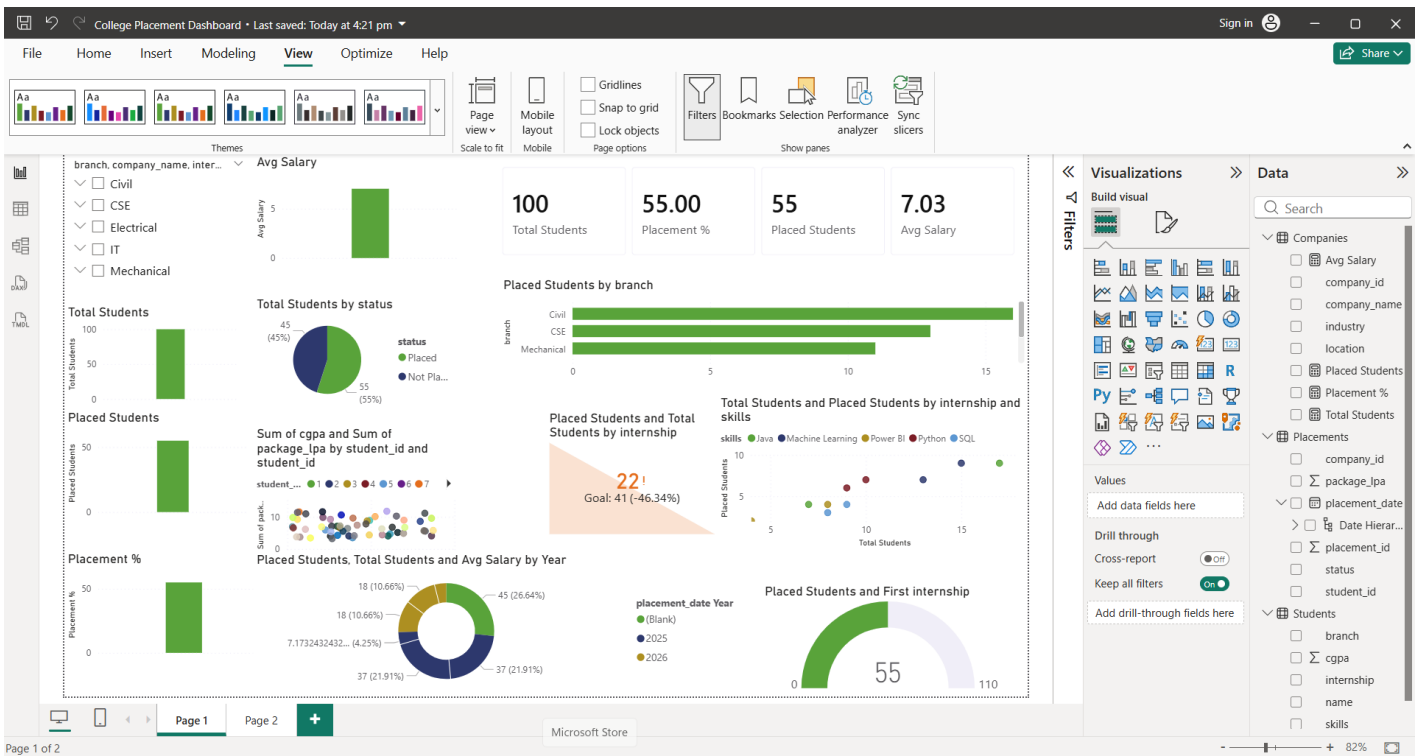
The College Placement Analysis Dashboard follows a structured data flow pipeline for transforming raw placement data into meaningful visual insights. The data is initially stored in an Excel dataset containing information about students, companies, and placement records.

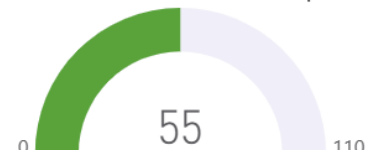
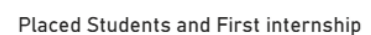
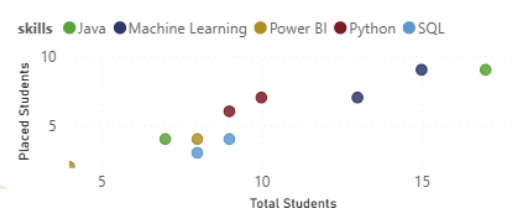
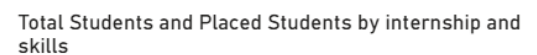
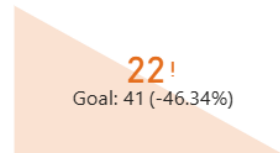
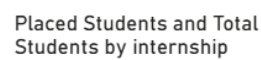
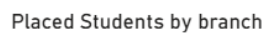
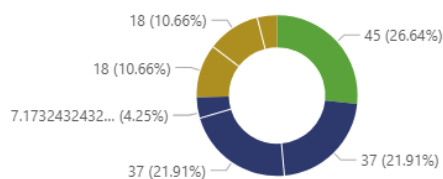
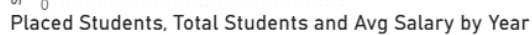
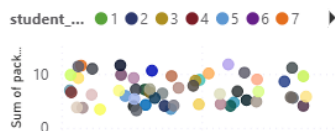
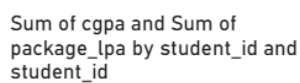
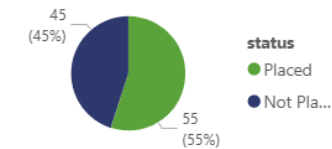
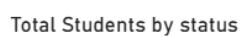
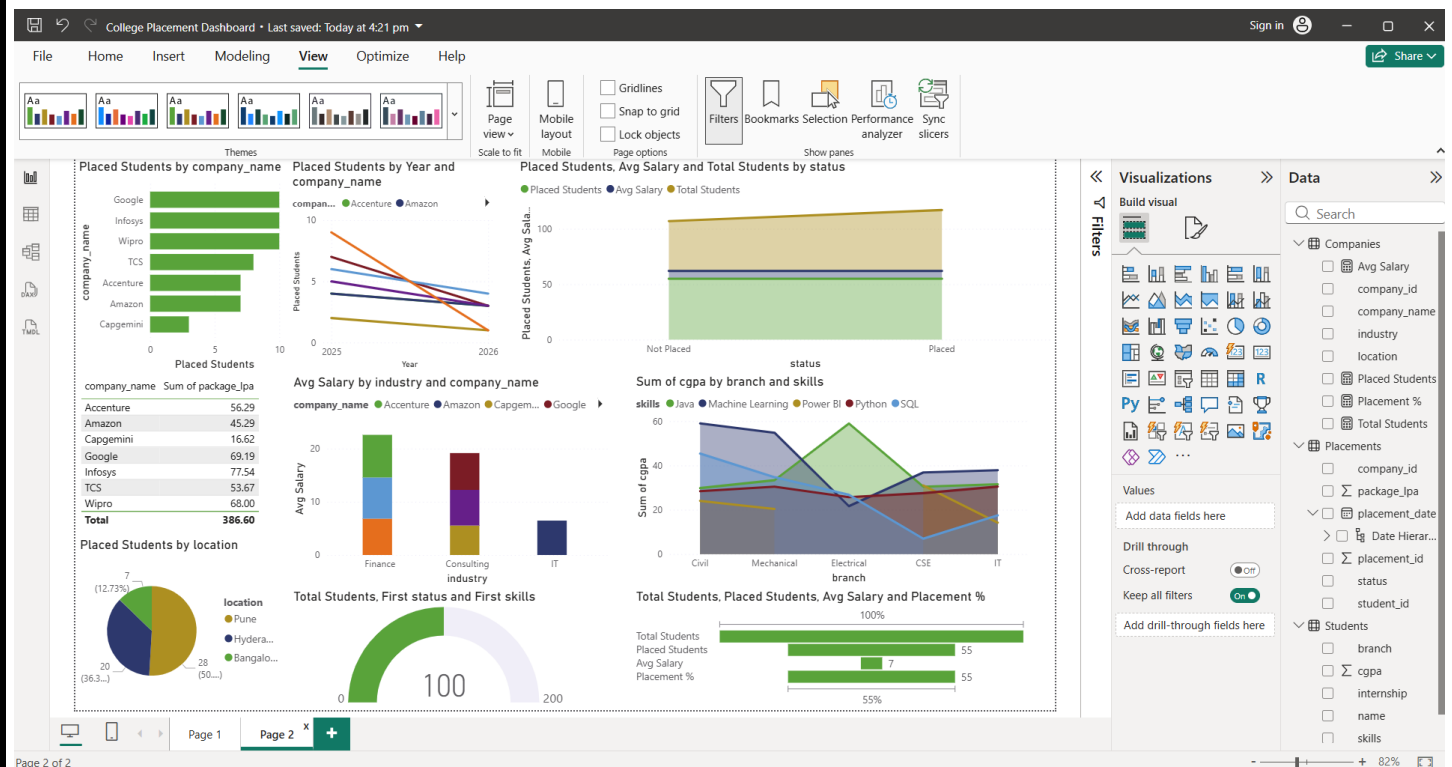
Using the **Power Query Editor**, the dataset is imported, cleaned, and transformed into a structured tabular format. Data preprocessing steps include handling missing values, renaming columns, and establishing relationships between tables such as Students, Companies, and Placements.

The transformed data is then loaded into the **Microsoft Power BI Desktop data model**, where relationships between multiple tables are defined to ensure accurate analysis. After data modeling, **DAX measures** are created to calculate key metrics such as total students, placed students, placement percentage, and average salary.

These calculated measures are then used to build interactive visuals on the dashboard canvas.

Screenshots of the project





5. Key Features

1. Data Integration

Placement data is imported from Excel and processed using Power Query Editor. This ensures clean, structured, and reliable data for analysis.

2. Placement Analysis

KPI cards display key metrics like Total Students, Placed Students, Placement %, and Avg Salary.

This provides a quick overview of overall placement performance.

3. Company & Salary Insights

The dashboard shows company-wise hiring and salary distribution. Users can identify top recruiters and high-paying companies easily.

4. Branch & Skills Analysis

Visuals analyze placement performance across different branches and skills. It helps understand which skills and branches have better placement outcomes.

5. Trend Analysis

Charts display year-wise placement and salary trends. This helps in identifying growth patterns and changes over time.

6. User-Friendly Design

The dashboard is designed with a clean layout and clear visuals. It improves readability and makes data easy to understand.

6. User Manual

Getting Started

- Open the College_Placement_Dashboard.pbix file in Microsoft Power BI Desktop.
- On first launch, go to Home > Transform Data to verify dataset and data model.
- Ensure the Excel dataset is properly connected and loaded.
- Click Refresh on the Home ribbon to load the latest placement data.

Navigating the Dashboard

- Use slicers (filters) to select Branch, Company, or Skills.
- View key placement metrics through KPI cards at the top.
- Navigate between Page 1 (Overview) and Page 2 (Company Analysis).
- Interact with charts to explore placement trends and insights.

Understanding the Visuals

- KPI Cards: Show Total Students, Placed Students, Placement %, Avg Salary.
- Bar Charts: Represent branch-wise and company-wise placements.
- Line/Area Charts: Show placement trends over years.
- Scatter Plot: Displays relationship between CGPA and salary.
- Gauge Chart: Shows placement performance summary

Data Refresh

- To manually refresh: Click Home > Refresh in Power BI.
- If dataset is updated in Excel, refresh to reflect latest data.
- For online use, publish and configure refresh in Power BI Service.

Troubleshooting

- If data is not visible: Check Excel file connection and path.
- If visuals appear blank: Click Refresh or clear filters.
- If dashboard is slow: Reduce visuals or check system performance.
- If file not opening: Ensure Power BI Desktop is installed/updated.
- If app is not working → Restart Power BI and check system.

7. Conclusion

The College Placement Analysis Dashboard successfully demonstrates the use of Microsoft Power BI Desktop for analyzing placement data and generating meaningful insights. It transforms raw data into interactive visualizations, helping users easily understand placement performance, company trends, and salary patterns.

The project achieves its objectives by providing clear insights through KPI cards, charts, and filters, making data analysis simple and effective. Overall, it highlights the importance of business intelligence tools in decision-making and serves as a strong foundation for future enhancements like predictive analytics and AI-based placement recommendations.

8. Links & References

API References

- OpenWeatherMap API: <https://openweathermap.org/api>
- WeatherAPI: <https://www.weatherapi.com/docs/>

Tool References

- Microsoft Power BI Documentation: <https://docs.microsoft.com/en-us/power-bi/>
- DAX Reference Guide: <https://dax.guide/>
- Power Query Documentation: <https://learn.microsoft.com/en-us/power-query/>